NETFLIX - ROOT CAUSE ANALYSIS

Problem Statement : You are the product manager at Netflix. You see there is a 10% drop in the Average View Duration. Find the Root Cause.

Defining Netflix?

Netflix, founded by Reed Hastings and Marc Randolph in 1997, started as a DVD rental service and later transformed into the world's leading subscription-based streaming platform.

It provides an extensive library of films, TV shows, documentaries, and exclusive Netflix Originals. The company operates in over 190 countries, offering localized content in multiple languages.

Netflix's unique recommendation algorithm personalizes content for each user, enhancing their viewing experience.

Over the years, it has reshaped the entertainment industry and maintained a strong influence in the global media landscape.

1. Clarifying Questions (To understand the context better)

1.1 User Segmentation:

1.1.1 Is the drop uniform across all user segments (regions, age groups, device types)?
Yes.

1.1.2 Are new or returning users more affected?
All of them .

1.1.3 Is there a specific geographic region where the drop is more pronounced?
No.

1.2 Content-Specific:

1.2.1 Is the drop specific to certain genres, shows, or movies?
All genres.

1.2.2 Are new releases underperforming compared to expectations?
Yes.

1.2.3 Did we change content recommendations recently?
Yes

1.3 Time Period:

1.3.1 When did we first notice the drop?
Approximately two weeks.

1.3.2 Is it seasonal (e.g., holidays, sports events affecting viewership)?
No such big event.

1.4 Platform/Device:

1.4.1 Is the drop consistent across devices (mobile, smart TV, web)?
Yes , All platforms

1.4.2 Have there been any recent UI/UX changes on specific platforms?
No such changes.

1.5 User Behavior:

1.5.1 Are users abandoning content at specific timestamps?
No.

1.5.2 Is there an increase in content switching or exits?
No.

1.5.3 Have there been changes in binge-watching behaviour?
No.

1.5.4 Has completion rates for episodes have dropped or users dropping off earlier?

Yes, completion rates for episodes have dropped, especially in the middle of the series and many users not watching.

1.6 Definition

1.6.1 How the AVD is calculated is it time duration or one active view per user? the definition of average view duration here (for e.g. if a user watches the same content multiple times in a single session, would each play be counted separately for Average View Duration, or is it treated as a single play?

Each play would be counted separately for Average View Duration, so if a user watches the same content multiple times, those would each count as individual plays in the calculation

1.6.2 10% drop as calculating all user's views taking into consideration or calculating average view per customer.
Yes , we are considering the views across the genres.

2. Sanity Check Questions (To rule out internal issues or anomalies)

2.1 Technical Issues:

2.1.1 Are there app crashes, buffering issues, or performance slowdowns?
There are few issues reported but not in a big number.

2.1.2 Any recent infrastructure changes that could impact streaming quality?
No.

2.1.3 Have CDN issues affected content loading times?
No.

2.2 A/B Testing and product :

2.2.1 Were there any experiments or A/B tests deployed recently?
Yes, there were some new modifications in the recommendation algorithm.

2.2.2 Are test groups experiencing different view duration patterns?
No.

2.2.3 Has the recommendation algorithm changed recently?
Yes.

2.3 UI/UX Changes:

2.3.1 Have there been any layout, autoplay, or navigation changes that could impact engagement?
No.

2.4 Pricing and Subscription:

2.4.1 Have there been changes in subscription plans, promotions, or price hikes that may affect viewing patterns?
No.

2.4.2 Are users cancelling or downgrading their plans more frequently?
Yes , 5% users downgraded or did not subscribe further.

2.5 Content Availability:

2.5.1 Have any popular show/movies been removed recently?
Yes , Popular shows like FRIENDS dropped from the Netflix subscription.

2.5.2 Are there licensing issues causing unexpected content unavailability?
No.

3. Pivot Questions (If the initial hypotheses don't reveal the root cause)

3.1 Competitor Landscape:

3.1.1 Are competitors (Disney+, Prime Video, etc.) launching compelling new content?
Yes, Disney+ and Amazon Prime released major movie ad series, which likely contributed to the shift in viewer attention.
.

3.1.2 Are users spending more time on alternative platforms?
No.

3.1.3 Is there any new product launched in the market?
Yes, Warner Bros launched its own streaming service as part of global strategy.

3.2 Social Media Trends:

3.2.1 Are negative reviews or social media feedback influencing user sentiment?
No.

3.2.2 Is there a viral trend affecting viewing habits?
No.

3.3 Market Trends:

3.3.1 Are there industry-wide changes (e.g., shorter attention spans, shift to short-form content like YouTube Shorts)?
Yes, attention spans are decreasing.

3.4User Surveys/Feedback:

3.4.1 Have there been complaints or concerns raised in customer support channels?
Yes, very frequent recent complaints regarding the issues of removing popular shows from Netflix.

3.4.2 Are there qualitative insights indicating dissatisfaction with content quality?
Yes, No change.

3.4.3 Are there any other issues came when doing the user servey?
Yes , users are opting for free content in various piracy sites and social media channels.

4. External factors -

4.1.1 Is there any Political events like elections or any political events are going on due to that any restrictions imposed?

- No

4.1.2 Is there any new entry of a similar product or competitor or something exclusive offering they are providing?

- Yes one major competitor released its streaming service as MAX and there was a business deal between Jio cinema of reliance industries in india i.e . Disney+ Hotstar.

4.1.3 Any kind of backlash facing on social media like on Twitter or Facebook due to any change in policy?

- No

4.1.4. Is there any recent regulatory announcement or policy affecting the industry? (Example: Stricter data privacy laws like GDPR enforcement in Europe)

- No

4.1.5. Are there any ongoing cultural or national holidays that could be diverting audiences towards outdoor or family activities? (Example: Thanksgiving in the U.S. or Chinese New Year)

- No

4.1.6. Any environmental factor? As I heard about the recent Los Angeles wildfires that devastated multiple homes in the Western United States—could that have contributed to the drop in average view duration?

- Yes, there were a few wildfires in the Western United States and typhoons in south east Asia.

5. Internal Factors

5.1.1 Is there any changes in the content repository?

Yes some very popular web-series are now not in the Netflix

5.1.2 Is there new content added?

Yes, there are various new contents added

5.1.3 What is the customer feedback of the new web series or content?

Most of the newly added web series are not getting good feedback fom the viewers.

5.1.4 Are there any changes in the average time spent on various pages?

Yes, viewers are spending more time on the landing page and with the search options.

5.1.5 Is there any issue reported as customers are having issues with the content on the page?

Yes, there are multiple issues reported across the world i.e. viewers are having issues finding their view-like content.

User journey analysis-

1. **User opens the Netflix app / Web app / Web site**
2. **User specific content with recommendation and last watched view.**
3. **Browsing for other content /newly added content**
4. **Selecting the content and play**
5. **Feedback of the content**

Root cause Hypothesis-

1. **Popular shows leaving Netflix as a platform –** Popular shows like FRIENDS left the Netflix platform.
2. **Competition in the market -** Launch of MAX streaming service worldwide and in the Indian market Disney and Jio cinema (Reliance) launched an integrated streaming platform.
3. **Changes in the recommendation algorithm-** There are changes in the recommendation algorithm that increased the search time of personalized content.
4. **Attention span-** Attention span is decreasing as users leave the shows which are having long streaming time.
5. **Environmental calamity-**There were various environmental disasters around the world.

Proposed solutions-

Short term solutions-

1. **Product Content Improvement:**

    1. As attention span is decreasing day by day Netflix needs to fragment the web series into more further divisions.(from 1 hour to 45 mins)
    2. Accelerate the release of popular, high-demand titles in the short term.
    3. Diversify the catalog to include trending genres and formats.
    4. Use data to identify underperforming content categories and adjust accordingly.

2. **Competitive business strategy :**

    1. Marketing campaigns using reels and short videos across the various social media platform.
    2. Launch aggressive marketing campaigns to highlight Netflix Originals and upcoming releases.
    3. Offer incentives like sneak peeks or early access to premium content and **new release** for loyal subscribers.

3. **Technical improvements:**

    1. Address any technical issues that might disrupt viewing.
    2. Optimize recommendation algorithms to surface highly relevant content for users with personalization.

Long term Solutions-

1. **Enhancing user experience-**

Start production of web series for new age audience which maps with the current trend.

2. **Gamification for user engagement –**

Gamification of the content as the user will be hooked to the content and watch introducing a Leaderboard of movie experts with question and answers

3. **Exclusive Netflix fan club –**

Introducing a Netflix fan club worldwide, users with having top leaderboard or top fan badge will have direct entry to the Fan event and meet their favorite celebrities.

4. **Netflix Coins-**
   Users will get coins as per the duration of the view time , coins which they can use to get a discount on their next subscription.
5. **Strategic partnership-**
   strategic partnership with producers for exclusive digital rights.
6. **Free Content –**

Provide unpopular shows as free content and earn money from advertisements.

7. **AI agent deployment-**
   AI agent deployment is needed to find pirated copies of Netflix's exclusive web series and movies and take initial legal steps.